

Multiple Choice (10 marks)

1. In Python 3, which of the following function removes all leading and trailing whitespace in string?
 - (a) `replace(old, new [, max])`
 - (b) `strip([chars])`
 - (c) `swapcase()`
 - (d) `title()`
2. Let `l = [1,2,3,4]`. What is `min(l)` in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
3. Which is the smallest asymptotically?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$
4. Which of the following best explains what happens when a new device is connected to the Internet?
 - (a) A device driver is assigned to the device.
 - (b) An Internet Protocol (IP) address is assigned to the device.
 - (c) A packet number is assigned to the device.
 - (d) A Web site is assigned to the device.
5. Which is the largest asymptotically?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$

True / False (10 marks)

Answer True or False

6. True or False: The sum of the list $l = [1, 2, 3, 4]$ in Python 3 is 10.
7. True or False: In Python 3, the expression $x \ll 3$ evaluates to 8 when x is 1.
8. True or False: In Python 3, the expression `['Hi!'] * 4` produces the list `['Hi!', 'Hi!', 'Hi!', 'Hi!']`.
9. True or False: A function that averages numeric values in a column or row is the best tool for detecting data entry errors in a dataset.
10. True or False: $O(\log n)$ grows faster than $O(n^2)$ for large input sizes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate the sum of series $1^2 + 2^2 + 3^2 + \dots + n^2$.
12. Write a function to display sign of the chinese zodiac for given year.

End of Paper